

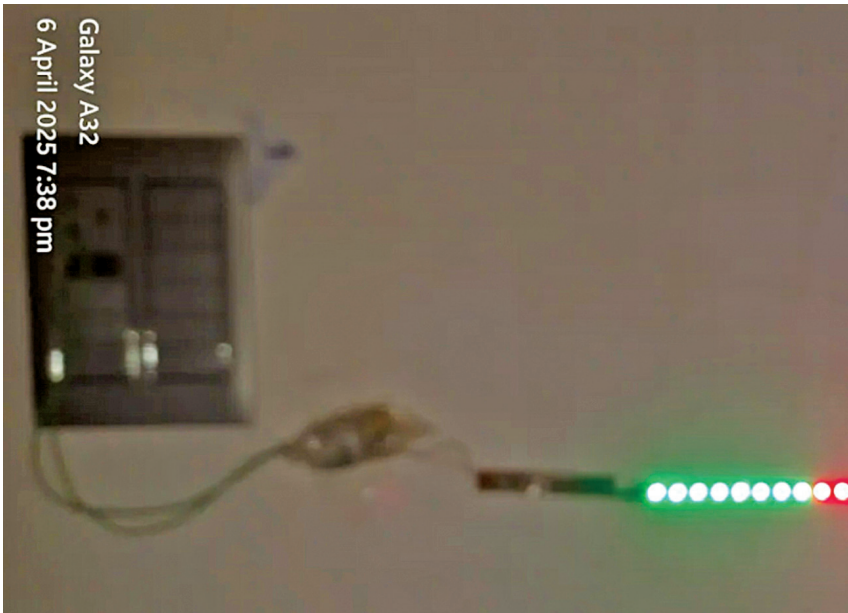


Fig. 4: Author's prototype with an LED strip

- Connect Arduino pin 10 to ATtiny85 pin 1 (Reset)
- Connect Arduino GND to ATtiny85 pin 4

Finally, place a 10 μ F capacitor between the GND and reset pins of the Arduino to disable auto-reset during programming. Once these steps are complete, the setup is ready to program the ATtiny85 chip using the Arduino.

Some modifications are required in the Arduino IDE to enable programming of the ATtiny85 chip. Follow these steps:

1. Open the Arduino IDE
2. Go to File > Preferences, and in the 'Additional Board Manager URLs' field, paste the following URL to add support for ATtiny boards:

https://raw.githubusercontent.com/damellis/attiny/ide-1.6.x-boards-manager/package_damellis_attiny_index.json

3. Click OK to save the settings
4. Next, go to Tools > Board > Boards Manager, search for 'attiny,' and install 'ATtiny by David A. Mellis'

5. From the Tools menu, set the following options:

- Programmer: Select Arduino as ISP
- Board: Select ATtiny25/45/85
- Processor: Choose ATtiny85
- Clock: Select Internal 16MHz
- Port: Choose the COM port to which the Arduino board is connected

The bootloader can now be burned, and sketches (software) can be uploaded to the ATtiny85.

Go to File > Examples > Basics, and select BareMinimum. Then, from the Tools menu, click Burn Bootloader. Once the bootloader has been successfully burned, the BareMinimum sketch (or any other sketch) can be uploaded to the ATtiny85.

In addition, the Adafruit NeoPixel library must be installed in the Arduino IDE. To do this,

TO DOWNLOAD SOURCE CODE AND RESOURCES



Scan this QR code or go to <https://sourcecode.electronicshub.in> to access and download source code and other resources for this project

go to Sketch > Include Library > Manage Libraries, then search for NeoPixel. Select 'Adafruit NeoPixel' by Adafruit and click Install.

Once the library is installed, upload the attached code for the profile lighting chasing effect to the ATtiny85 chip by clicking the upload button, or go to Sketch > Upload using programmer. The setup is now complete.

Construction and testing

After the code has been successfully uploaded, unplug the Arduino and carefully remove the ATtiny85 from the breadboard. Then, insert the ATtiny85 into the 8-pin IC base on the controller circuit.

Operation is straightforward. Once the device is powered, bringing a finger near the micro-wave radar sensor triggers the system. When a finger is placed near the sensor—or someone walks close to it—the radar detects motion. In response, the LED strip gradually lights up, turning on each LED sequentially in a warm colour. As long as motion is detected, the strip cycles through various warm colours every second (such as orange, gold, pink, etc), creating a cosy and inviting ambience.

If no motion is detected for 10 seconds, the LEDs begin to turn off one by one in reverse order, producing a smooth fade-out effect. This setup is ideal for hallway lighting, bedroom ambience, or decorative installations that respond to movement.

The controller features ambient colour chasing, allowing both the colour scheme and the speed of transitions to be customised through programming. **EFY**



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